

VAHNIM · GROUNDNODE

Simulation User Manual

How to run and present the city simulation and the Ground-N1 aircraft-to-ground demo: three windows, one shared simulation, five pads.

Covers: City simulation (home) · Ground-N1 launcher, pilot screen, ground control, live 3D window · Model hangar · Smart pad · Cockpit encounter · Integration lab

1. What this is

The project simulates an urban air mobility terminal with **five landing pads (GN-01 to GN-05)**. It has two parts you can use on their own or together:

- **City simulation (home page).** A fleet of UAMs (passenger aircraft) and UAVs (small drones) flies out from the pads, hovers over a city site, returns, lands, charges and redeploys. Use it to show fleet scale and traffic separation.
- **Ground-N1 demo.** One UAM talks to the ground node. The pilot uses a cockpit screen, the ground-control team uses an operations screen, and a live 3D window shows the pad and the aircraft. Every instruction is a signed, verified datalink message and the ground automation explains its decisions.

Contents

1. What this is
2. Getting started
3. Navigating between pages
4. City simulation (home)
5. Ground-N1 demo: launcher and windows
6. Ground-N1 walkthrough (step by step)
7. Scenarios, supervision and tests
8. Other pages
9. Troubleshooting and limits

2. Getting started

1. Install **Node.js** (any recent version) and open a terminal in the project folder.
2. Run `npm install` once. This downloads the pinned 3D library (three.js).
3. Run `npm start`. It serves the project at `http://localhost:8123`.
4. Open `http://localhost:8123` in **Chrome or Edge**. The home page is the city simulation.

Check it works: run `npm test`. All suites should print PASS lines (city fleet, protocol, coordination, encounter and Ground-N1).

Requirements

- A desktop browser with WebGL (Chrome, Edge, Firefox). The three Ground-N1 windows must be in the **same browser** so they can share one simulation.
- For the two-screen demo use a display of at least 1280 × 720 per window. A second monitor is ideal for the live window.

Page map

Page	File	Purpose
City simulation (home)	index.html	Whole fleet, five pads, city missions
Ground-N1 launcher	n1.html	Choose scenario, mode and speed; open the windows
Pilot	n1-pilot.html	Aircraft cockpit software
Ground control	n1-ground.html	Operations screen for the ground team
Live 3D	n1-live.html	3D view of the terminal and aircraft
Demo wall	n1-wall.html	All three screens tiled on one display
Explore overview	explore.html	Plain-language menu of the pages below, with a glossary
Cockpit encounter	comms.html	Guided practice landing with a coach (private simulation)
Aircraft models	hangar.html	Turntable viewer with hover, cruise and parked modes and key facts
Smart pad	smart-pad.html	Interactive pad concept
Integration lab	coordination-lab.html	Detailed review and failure exercises

3. Navigating between pages

Every page has the same bar across the top. The **orange** ← **Home** button (left) returns to the city simulation from any page. The other tabs are grouped as **Ground-N1** (Launcher, Pilot, Ground control, Live 3D, Demo wall) and **Explore** (Overview, Cockpit encounter, Aircraft models, Smart pad, Integration lab). The current page is highlighted. On narrow windows the bar scrolls sideways.

At the right end of the bar, the pulsing ⓘ **In simple words** button opens a short plain-language explanation of the page you are on. Click *Got it* to close it; the button stops pulsing once you have used it.

Clicking a tab replaces the current window. During a Ground-N1 demo keep the Pilot, Ground and Live windows open; the simulation keeps running while at least one of them is open, and restarts if you close all of them.

4. City simulation (home)

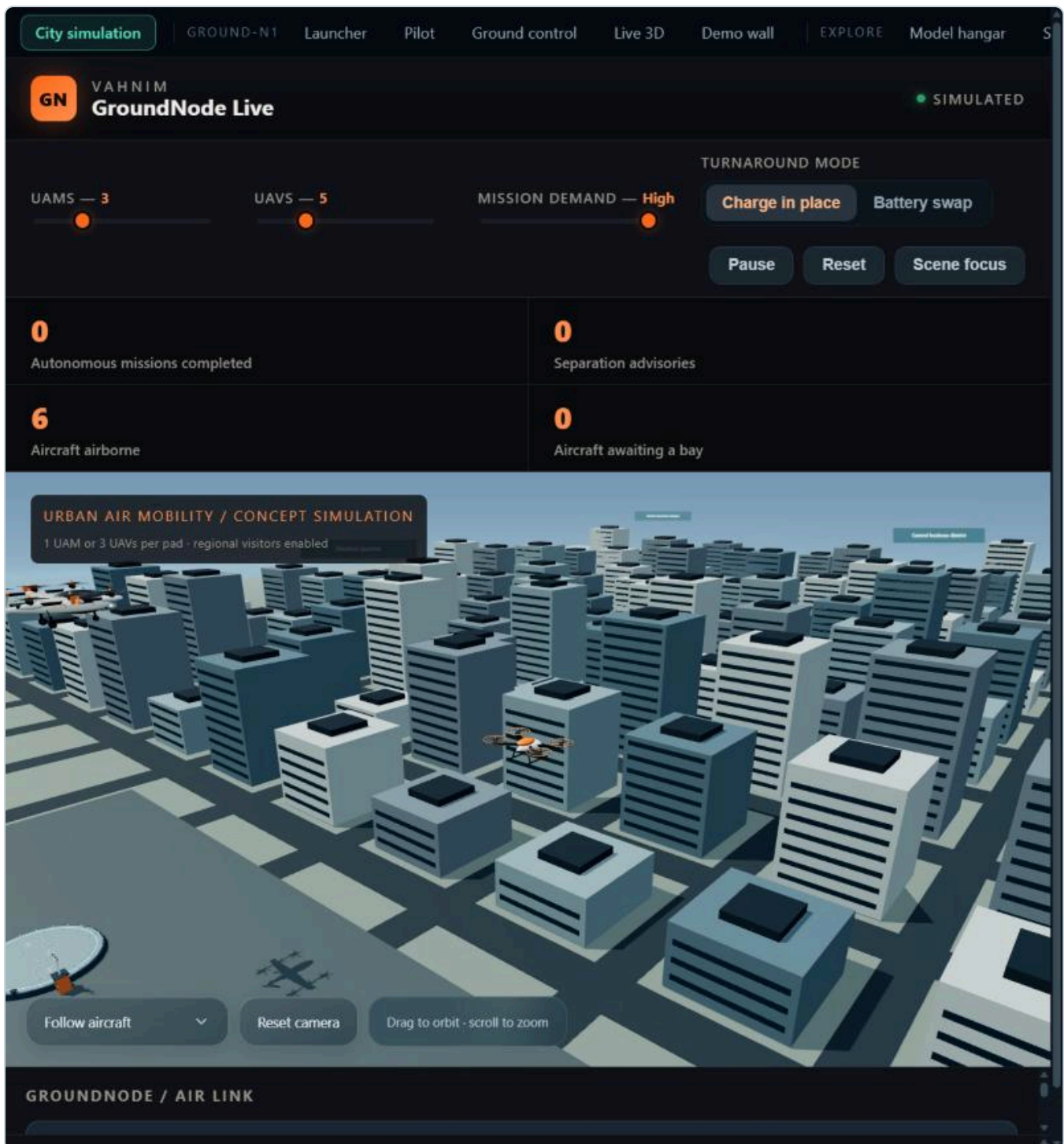


Figure 1. City simulation: controls (top), 3D scene with the five pads, flight board and communications.

What you see

Aircraft depart a pad vertically, climb, fly to a city site, **hover still** for the mission, then return, wait for a free bay, land, charge (or swap the battery) and redeploy. Each UAM takes a whole pad. Each pad holds up to three UAVs, never mixed with a UAM.

Controls

Control	What it does
UAMs slider (0 to 12)	Number of passenger aircraft in the home fleet. Added aircraft enter from outside the city; removed ones retire after they land and finish service.
UAVs slider (0 to 20)	Number of drones in the home fleet.
Mission demand	How often idle aircraft are dispatched. Higher demand means busier pads and queues.
Turnaround mode	<i>Charge in place</i> (longer dwell) or <i>Battery swap</i> (short dwell).
Pause / Reset	Freeze the simulation (camera still works) or restart it.
Scene focus	Hides the panels so only the 3D scene shows.
Camera menu	<i>Orbit / overview</i> , <i>City overview</i> , <i>Follow aircraft</i> (tracks the selected aircraft), <i>Pad approach</i> (close on the selected pad). Drag to orbit, scroll to zoom, <i>Reset camera</i> to go home.
Flight board	Click an aircraft to select it and see mission, battery, altitude and traffic status.
Communications filter	Show all channels, missions, separation advisories, city handoffs, arrivals, departures, service, or only the selected aircraft or pad.

The four counters show missions completed, separation advisories, aircraft airborne and aircraft waiting for a bay. Watch "waiting for a bay" rise when you raise demand with few pads free.

5. Ground-N1 demo: launcher and windows

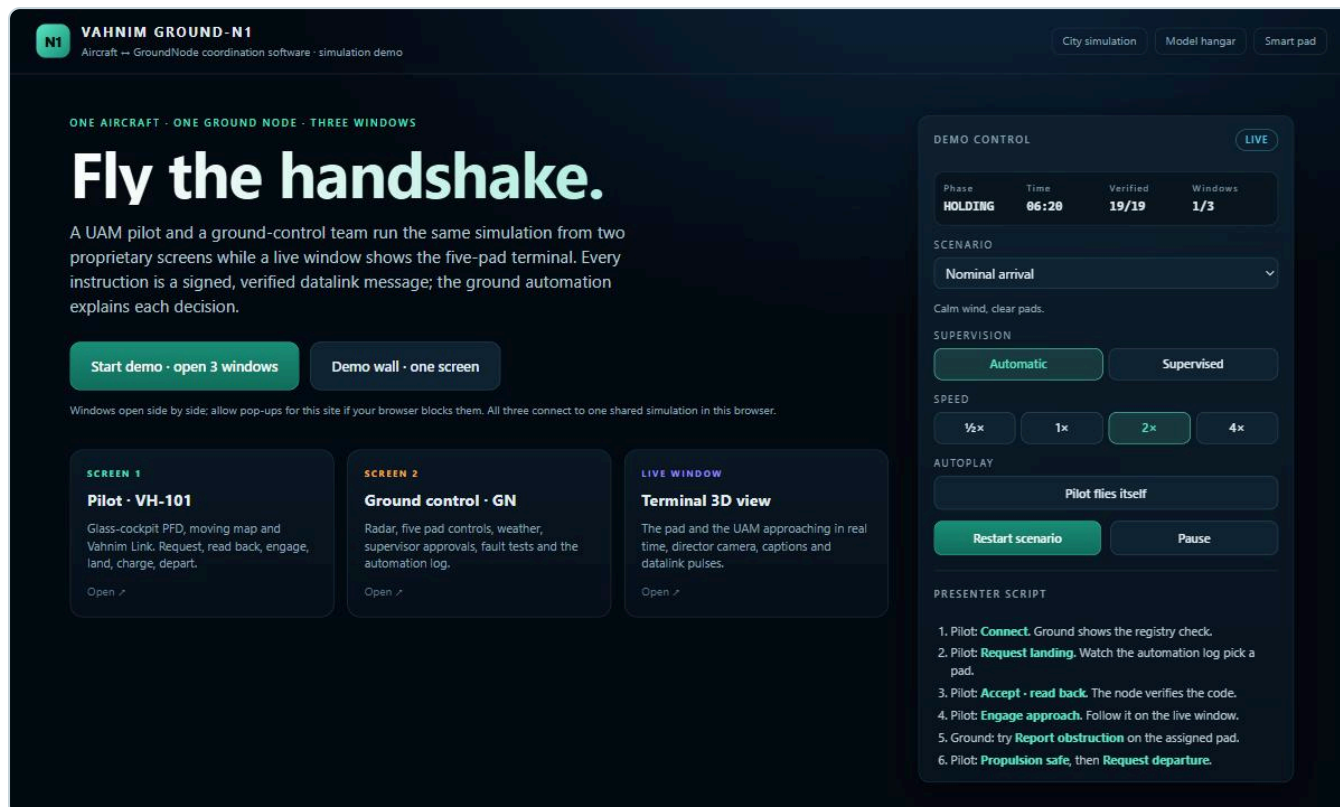


Figure 2. Launcher (Ground-N1 > Launcher). Demo control panel on the right.

Starting the demo

1. Open **Ground-N1 > Launcher**.
2. Pick a **Scenario**, a **Supervision** mode and a **Speed** (1/2x, 1x, 2x, 4x).
3. Click **Start demo · open 3 windows** (allow pop-ups). Live fills the screen; Pilot and Ground share the screen or go to a second display. Or click **Demo wall** to see all three on one display.

The panel shows the aircraft phase, simulated time, verified message count and how many windows are connected. **Restart scenario** resets everything; **Pause** freezes flight and timers; **Autoplay** lets the pilot press every recommended step by itself.

The three windows

Window	Who	Main elements
Pilot	UAM pilot	Flight display (attitude, speed, altitude, vertical speed, heading), moving map, mission progress bar, clearance card with read-back code, action buttons, verified datalink log.
Ground control	Ground team	Terminal radar, traffic strip, five pad cards with controls, weather sliders, supervision toggle, approval card, intervention and test buttons, datalink log, automation decision log.
Live 3D	Everyone	The five-pad terminal and the aircraft. Director camera, captions for key events, telemetry, message ticker.

Pilot screen

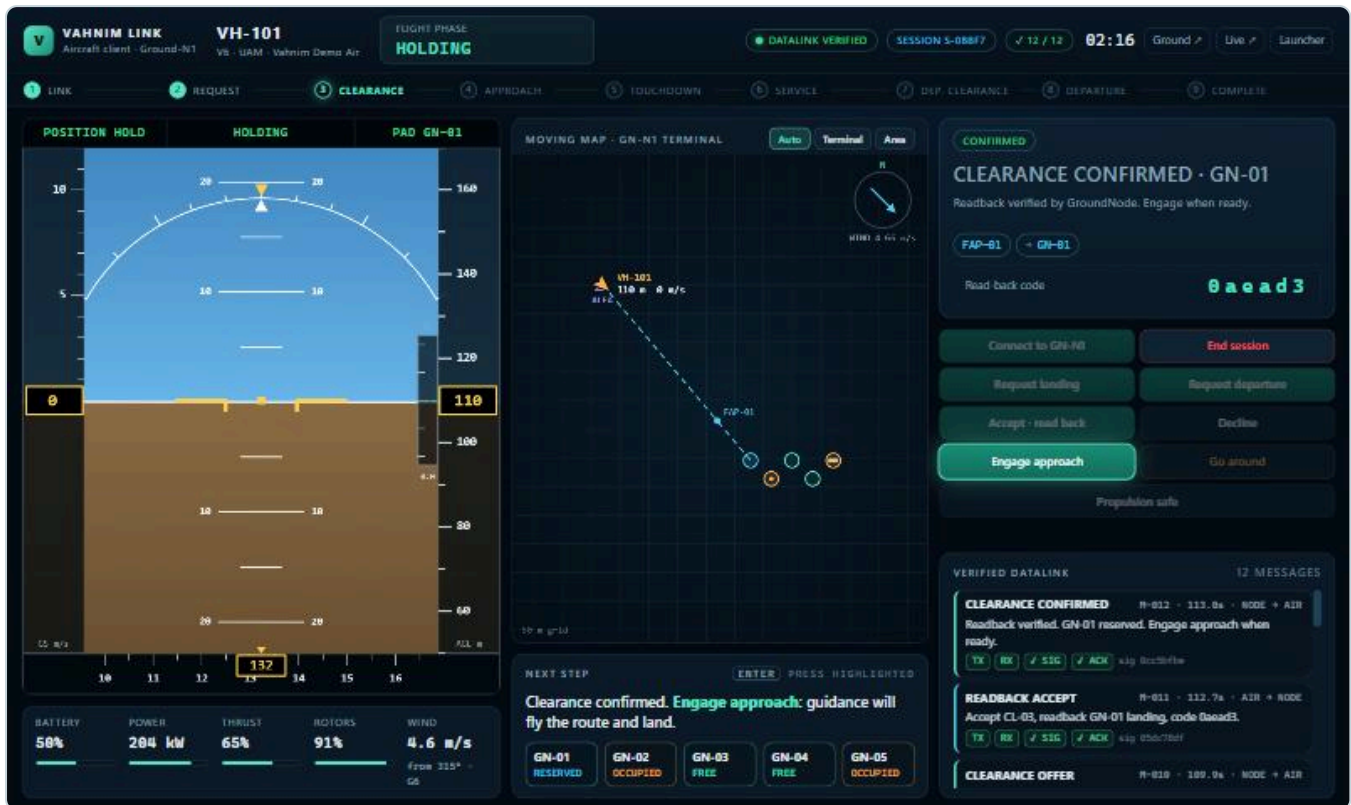


Figure 3. Pilot screen after a clearance is confirmed. The highlighted button is the next step.

Button	Use
Connect to GN-N1	Open a verified session; the node checks the aircraft against its registry.
Request landing / Request departure	Ask the ground node. It answers automatically.
Accept · read back / Decline	Answer a clearance offer. Accepting sends the code back for the node to verify. Offers expire after 30 s.
Engage approach / Engage departure	Starts the guidance. Accepting alone never moves the aircraft.
Go around	Abort the approach: climb and hold.
Propulsion safe	After landing and rotor stop; starts ground power and charging.
End session	Closes the datalink session.

Keyboard: **Enter** presses the highlighted next step, **G** commands a go-around. The map has **Auto / Terminal / Area** range buttons.

Ground control screen

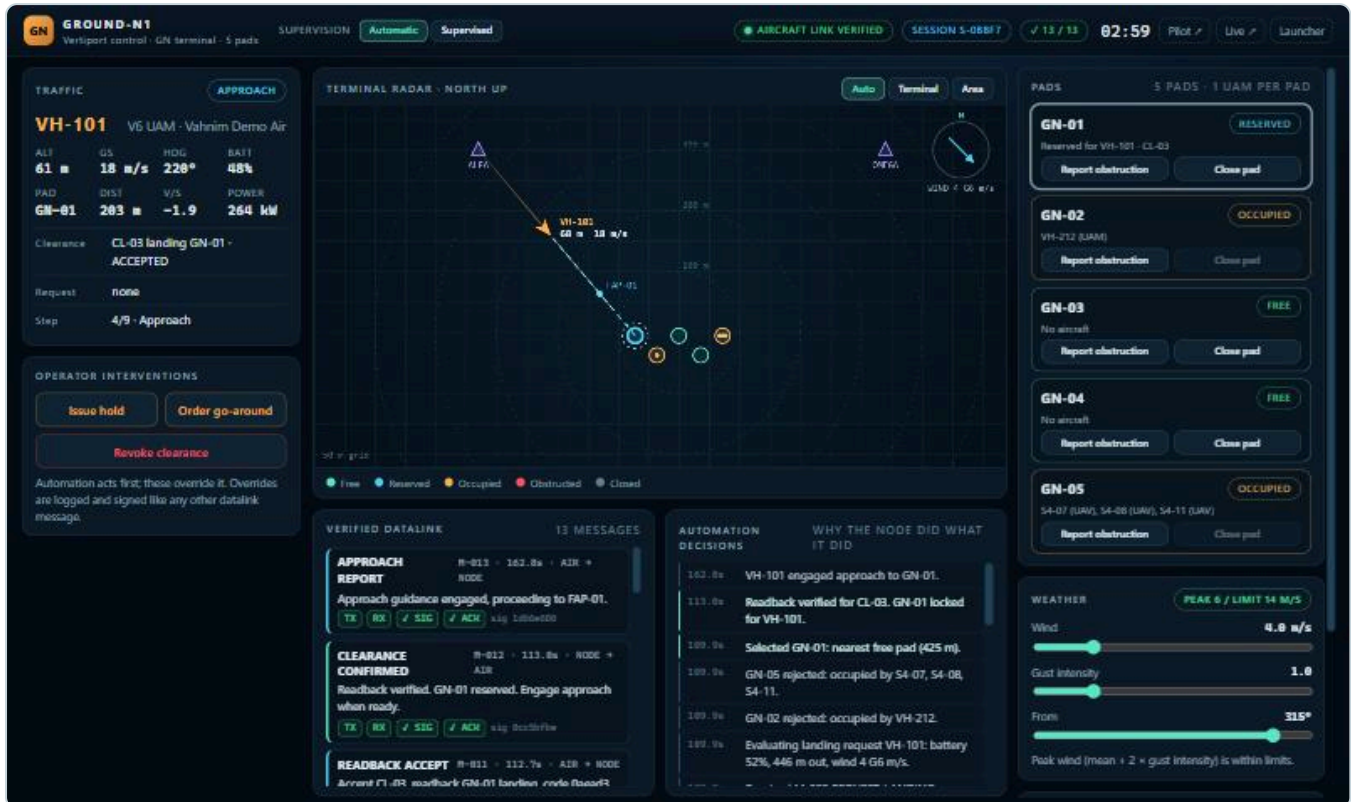


Figure 4. Ground control during an approach to GN-01: radar, five pads, weather and both logs.

Element	Use
Supervision (top)	Automatic: the node offers clearances at once. Supervised: each recommendation waits for an operator.
Approval card	Appears in Supervised mode. Approve the recommended pad, pick another free pad, or Deny to hold.
Pad cards	<i>Report obstruction</i> (debris on a pad) and <i>Close pad</i> (maintenance). Both remove the pad from use; an active approach to it is revoked and diverted.
Weather	Wind speed, gust intensity and direction. When the peak wind passes the limit (14 m/s) requests are held.
Interventions	Issue hold, order go-around, revoke clearance. They override automation and are logged.
Verification tests	Send forged message, replay last message, drop the datalink for 10 s.
Automation decisions	Plain-language reasons: why a pad was chosen or rejected, holds and diverts.

Live 3D window

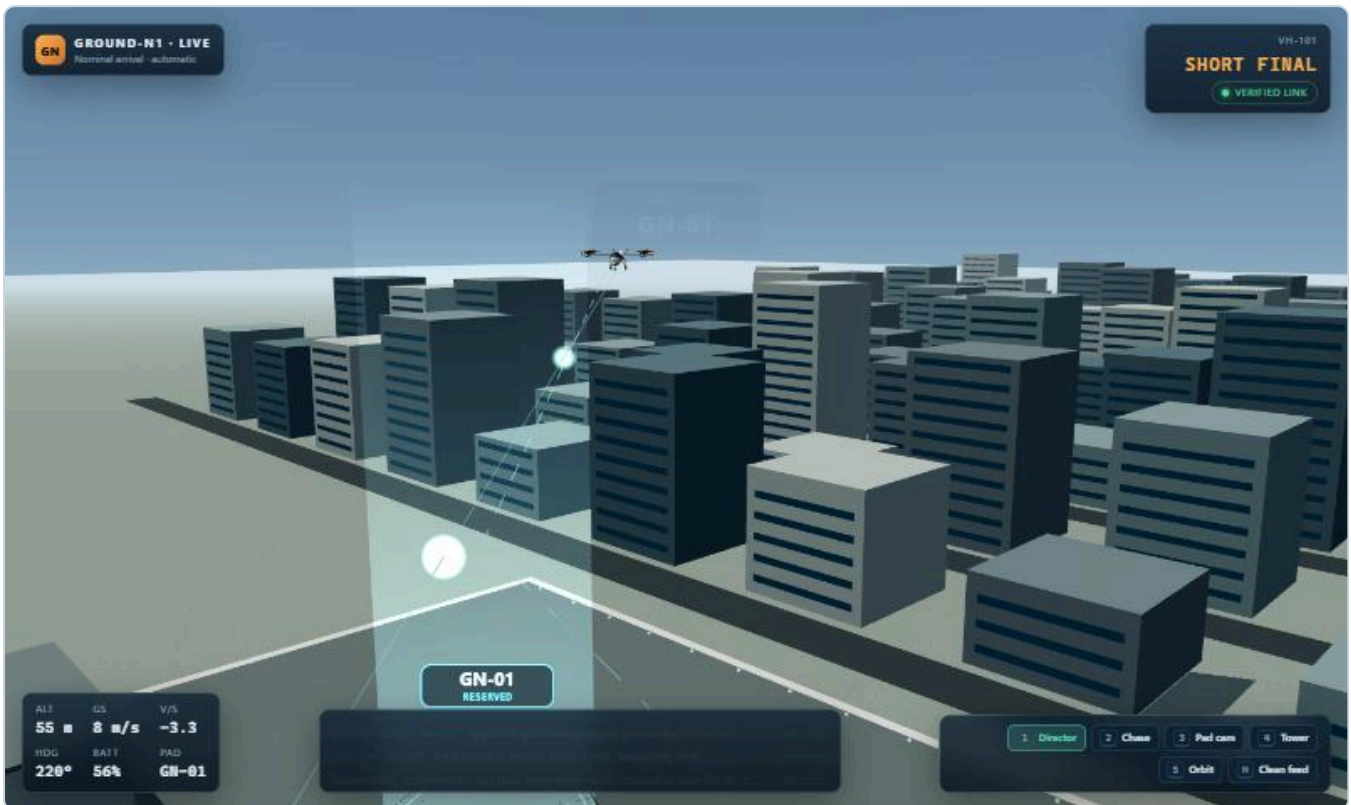


Figure 5. Live window on short final: the beam marks the assigned pad, labels show pad status.

Pad rings and labels follow status: **green** free, **cyan** reserved, **amber** occupied, **red** obstructed, **grey** closed.

Cameras: **1** Director (automatic shots), **2** Chase, **3** Pad cam, **4** Tower, **5** Orbit (drag and scroll), **6** Cockpit seat (first-person from the windshield). **H** hides the panels and the top bar for a clean feed, **F** toggles fullscreen.

6. Ground-N1 walkthrough (step by step)

Use the **Nominal arrival** scenario at 1×. A full run takes about two and a half minutes. Progress shows in the bar under the pilot header.

#	Who	Action	What happens
1	Pilot	Connect	Signed HELLO; the node verifies the identity and replies with the session.
2	Pilot	Request landing	The automation checks all five pads, wind and battery. Occupied pads are rejected with the reason; it picks the nearest free pad.
3	Pilot	Accept · read back	The read-back code is verified; the pad is locked and shows as reserved.
4	Pilot	Engage approach	The aircraft flies the route, aligns over the pad and descends. Watch the live window.
5	Automatic	Touchdown	The touchdown report and sink rate are verified. Rotors spool down.
6	Pilot	Propulsion safe	Charging starts (time-compressed for the demo).
7	Pilot	Request departure	Offered once battery and weather allow; accept, then Engage departure .
8	Automatic	Climb-out	The pad is released when the aircraft is clear; the session closes.

Reading a message

Each datalink entry shows kind, time, direction and text, plus chips: **TX** sent, **RX** delivered, ✓ **SIG** signature verified, ✓ **ACK** acknowledged. A rejected message is red with the reason.

Signing in the demo uses a keyed hash to show verification. It is not real cryptography.

7. Scenarios, supervision and tests

Scenario	What to show
Nominal arrival	The full happy path.
Gusty wind hold	The request is held for wind. On the ground screen lower the wind slider; the automation reassesses and offers a clearance by itself.
Pad incursion drill	Debris appears on the assigned pad on short final: clearance revoked, go-around, divert offer to another pad.
Low battery priority	Battery below reserve: the request is flagged priority and skips supervisor approval.
Link-loss drill	The datalink drops on approach: the aircraft goes around and holds. When the link returns, Connect again and re-request.

Things to try during any run

- Switch to **Supervised** and approve a different pad than the one recommended.
- On the ground screen, **Report obstruction** on the assigned pad mid-approach.

- Click **Send forged message** and **Replay last message**: both are rejected and nothing changes.
- Set the wind above the limit on final: the node orders a go-around.

Tips for presenting

- Use **Autoplay** with the Demo wall for a hands-free loop; use 2× or 4× to shorten waits.
- Put the Live window on the big screen and press H for a clean feed.

8. Other pages

Explore overview

A menu that explains each page in plain words (what it is, what to try, who it is for) plus a short glossary of terms such as UAM, clearance and go-around.

Cockpit encounter (guided)

The easiest way to understand the whole system. It runs its own private simulation, so it never disturbs a Ground-N1 demo running in other windows.

- **Coach (top):** a progress strip, *What is happening* in simple words, and *What to do next*. Below it, the last decision made by the ground computer.
- **Left:** a 3D view from the cockpit seat. Use the camera buttons (or keys 1 to 6) to switch to chase, pad, tower or orbit views.
- **Right:** the Pilot screen, or the Ground team screen using the tabs. It is the same software as the Ground-N1 windows.
- **Cause a problem (bottom):** *Block the pad*, *Strong wind*, *Drop the radio link*, *Send a fake message*, and *A person approves landings* (supervised mode). Press again to undo. The coach explains what the system does about each.
- **Top right:** *Watch it fly itself* (autoplay), *Speed* (1×, 2×, 4×) and *Restart*. The earlier version is still available through the *Classic version* link.

Aircraft models

Turntable view of the Vahnim V6 air taxi (UAM) and S4 drone (UAV) with plain-language key facts. Choose **Hover** (rotors lifting), **Cruise** (the air taxi's wing carries it and its lift rotors stop; the drone tilts forward) or **Parked**. Drag to look around; scroll to zoom.

Smart pad

Interactive 3D pad concept. Click a part or the system list to highlight it; use Top view, Reset, auto-rotate and the exploded view.

Integration lab

Detailed exercise for equipped and visitor aircraft with operator review and failure cases (expiry, conflicts, link loss). It can also mirror city telemetry from another tab.

9. Troubleshooting and limits

Problem	Fix
Page will not load	Check <code>npm start</code> is running and use <code>http://localhost:8123</code> .
Windows do not open	Allow pop-ups for the site, or open the pages from the tabs, or use the Demo wall.
Screens show different states	Windows must be in the same browser profile. Add <code>?local=1</code> to a page only if you want a private, unsynced copy.
Everything restarted	Closing every Ground-N1 window ends the shared simulation. Keep at least one open.
3D scene is blank	The browser needs WebGL. Try Chrome or Edge with hardware acceleration on.
Buttons are greyed out	Actions only enable when valid. Follow the highlighted step, or read the hint under "Next step".

Limits. This is a simulation for demonstration. Flight dynamics are illustrative, not validated aerodynamics. Only one aircraft is simulated in Ground-N1; other pads show parked aircraft. Charging is time-compressed. The pilot screen is a software concept, not a certified flight instrument. Registry, read-back and message signing are demonstrations. No aviation regulatory compliance is claimed.

Tuning notes for developers (pad positions, limits, scenarios, aircraft specifications, colours) are in the project README under "Tuning cheat-sheet".